

Relational Algebra

Tridib Banik, 400514461

November 6, 2025

1 Q1

$$\tau_{\text{NumComplaints DESC}} \left(\sigma_{\text{NumComplaints} > 2} \left(\gamma_{\text{TripID}, \text{COUNT}(*)} \rightarrow \text{NumComplaints}(\text{Complaints}) \right) \right)$$

2 Q2

$$\begin{aligned} J : \\ &= \rho_p(\text{Person}) \\ &\bowtie_{p.\text{PersonID}=d.\text{PersonID}} \rho_d(\text{Drivers}) \\ &\bowtie_{d.\text{PersonID}=t.\text{DriverID}} \rho_t(\text{Trips}) \\ &\bowtie_{t.\text{TripID}=c.\text{TripID}} \rho_c(\text{Complaints}) \end{aligned}$$

$$\begin{aligned} C_{\text{filters}} : \\ &= (c.\text{Category} \\ &= \text{'driver conduct'}) \wedge (c.\text{Timestamp} \\ &\geq \text{'2024-11-01'}) \wedge (c.\text{Timestamp} \\ &\leq \text{'2024-11-10'}) \end{aligned}$$

$$\tau_{p.\text{PersonID ASC}} \left(\delta(\pi_{p.\text{PersonID}, p.\text{Name}}(\sigma_{C_{\text{filters}}}(J))) \right)$$

3 Q3

$$\begin{aligned} J : \\ &= \rho_r(\text{Routes}) \\ &\bowtie_{r.\text{RouteID}=t.\text{RouteID}} \rho_t(\text{Trips}) \\ &\bowtie_{t.\text{TripID}=tk.\text{TripID}} \rho_{tk}(\text{Takes}) \\ &\bowtie_{tk.\text{RiderID}=ri.\text{PersonID}} \rho_{ri}(\text{Riders}) \end{aligned}$$

$$\begin{aligned} J_{\text{filtered}} : \\ &= \sigma_{tk.\text{Date} \geq \text{'2024-11-01'} \wedge tk.\text{Date} \leq \text{'2024-12-31'}}(J) \end{aligned}$$

$$\tau_{\text{NumRiders DESC}} \left(\gamma_{r.\text{RouteID}, r.\text{Name}, \text{COUNT}(\delta(ri.\text{PersonID}))} \rightarrow \text{NumRiders}(J_{\text{filtered}}) \right)$$

4 Q4

$$\begin{aligned} S_{\text{filtered}} : \\ = \sigma_{\text{SubwayStation}='STATION_M140'}(\rho_s(\text{Stops})) \end{aligned}$$

$$\begin{aligned} J : \\ = \rho_r(\text{Routes}) \\ \bowtie_{r.\text{RouteID}=t.\text{RouteID}} \rho_t(\text{Trips}) \\ \bowtie_{t.\text{TripID}=st.\text{TripID}} \rho_{st}(\text{Stoptimes}) \\ \bowtie_{st.\text{StopID}=s.\text{StopID}} S_{\text{filtered}} \end{aligned}$$

$$\delta(\pi_{r.\text{Name}}(J))$$

5 Q5

$$\begin{aligned} A : \\ = \pi_{\text{PersonID}}(\rho_{ri}(\text{Riders})) \end{aligned}$$

$$\begin{aligned} B_{\text{joined}} : \\ = \sigma_{tk.\text{Date} \geq '2024-12-01' \wedge tk.\text{Date} \leq '2024-12-31'}(\rho_{tk}(\text{Takes})) \\ \bowtie_{tk.\text{TripID}=t.\text{TripID}} \rho_t(\text{Trips}) \end{aligned}$$

$$\begin{aligned} B : \\ = \pi_{tk.\text{RiderID}}(B_{\text{joined}}) \end{aligned}$$

$$\tau_{\text{PersonID ASC}}(A - \rho_{\text{RiderID} \rightarrow \text{PersonID}}(B))$$

6 Q6

$$\begin{aligned} J : \\ = \rho_r(\text{Routes}) \\ \bowtie_{r.\text{RouteID}=t.\text{RouteID}} \rho_t(\text{Trips}) \\ \bowtie_{t.\text{ScheduleID}=s.\text{ScheduleID}} \rho_s(\text{ScheduleDays}) \end{aligned}$$

$$\begin{aligned} J_{\text{filtered}} : \\ = \sigma_{(s.\text{ScheduleBitmap} \ \& \ 5) \neq 5}(J) \end{aligned}$$

$$\begin{aligned} G : \\ = \gamma_{r.\text{Name}, \text{COUNT}(t.\text{TripID}) \rightarrow \text{Cnt}}(J_{\text{filtered}}) \end{aligned}$$

$$\delta(\pi_{r.\text{Name}}(\sigma_{\text{Cnt} \geq 1}(G)))$$

7 Q7

$$\begin{aligned}
J_{\text{joined}} : & \\
&= \rho_{p1}(\text{Person}) \\
&\bowtie_{p1.\text{PersonID}=ri.\text{PersonID}} \rho_{ri}(\text{Riders}) \\
&\bowtie_{ri.\text{PersonID}=c.\text{RiderID}} \rho_c(\text{Complaints}) \\
&\bowtie_{c.\text{TripID}=t.\text{TripID}} \rho_t(\text{Trips}) \\
&\bowtie_{t.\text{DriverID}=d.\text{PersonID}} \rho_d(\text{Drivers}) \\
&\bowtie_{d.\text{PersonID}=p2.\text{PersonID}} \rho_{p2}(\text{Person}) \\
&\bowtie_{t.\text{RouteID}=r.\text{RouteID}} \rho_r(\text{Routes})
\end{aligned}$$

$$\begin{aligned}
C_{\text{filters}} : & \\
&= (c.\text{Category} \\
&= \text{'overcrowding'}) \wedge (c.\text{Timestamp} \\
&\geq \text{'2024-12-18 00:00:00.000000'}) \wedge (c.\text{Timestamp} \\
&\leq \text{'2024-12-18 23:59:59.999999'}) \wedge (ri.\text{PersonID} \\
&= 1313)
\end{aligned}$$

$$\begin{aligned}
J : & \\
&= \sigma_{C_{\text{filters}}}(J_{\text{joined}})
\end{aligned}$$

$$\begin{aligned}
\pi_{p1.\text{Name} \rightarrow \text{RiderName}, c.\text{Timestamp} \rightarrow \text{ComplaintTime}, \quad (J) \\
c.\text{Category} \rightarrow \text{ComplaintCategory}, p2.\text{Name} \rightarrow \text{DriverName}, \\
d.\text{LicenseNo} \rightarrow \text{LicenseNo}, r.\text{Name} \rightarrow \text{RouteName}, \\
t.\text{Direction} \rightarrow \text{Direction}}
\end{aligned}$$

8 Q8

$$\begin{aligned}
J_{\text{joined}} : & \\
&= \rho_t(\text{Trips}) \\
&\bowtie_{t.\text{TripID}=tk.\text{TripID}} \rho_{tk}(\text{Takes}) \\
&\bowtie_{tk.\text{RiderID}=ri.\text{PersonID}} \rho_{ri}(\text{Riders}) \\
&\bowtie_{t.\text{DriverID}=p1.\text{PersonID}} \rho_{p1}(\text{Person}) \\
&\bowtie_{p1.\text{PersonID}=d.\text{PersonID}} \rho_d(\text{Drivers}). \\
&\bowtie_{t.\text{ScheduleID}=s.\text{ScheduleID}} \rho_s(\text{ScheduleDays})
\end{aligned}$$

$$\begin{aligned}
C_{\text{filters}} : & \\
&= (s.\text{StartDate} \\
&\leq \text{'20241231'}) \wedge (s.\text{EndDate} \\
&\geq \text{'20241028'}) \wedge (s.\text{ScheduleBitmap} \\
&= 124)
\end{aligned}$$

$$\begin{aligned}
J : & \\
&= \sigma_{C_{\text{filters}}}(J_{\text{joined}})
\end{aligned}$$

$$\begin{aligned}
G : \\
= \gamma_{p1.Name, \text{COUNT}(\delta(t.TripID)) \rightarrow \text{TotalTrips}, \text{COUNT}(tk.RiderID) \rightarrow \text{TotalPassengers}}(J)
\end{aligned}$$

$$\begin{aligned}
H : \\
= \sigma_{\text{TotalPassengers} \geq 800}(G)
\end{aligned}$$

$$\tau_{\text{TotalTrips DESC}} \left(\pi_{p1.Name \rightarrow \text{DriverName}, \text{TotalTrips}, \text{TotalPassengers}}(H) \right)$$